DATA ANALYTICS CAPSTONE PROJECT

Customer Sentiment Analysis on Flipkart Product Reviews

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Internship Domain

Data Analytics

Submitted To

LaunchED Global Internship

College

GITAM (Deemed to Be) University

Course

Computer Science and Engineering (Data Science)

Academic Year

2025–2026

ACKNOWLEDGEMENT

Acknowledgement

I would like to express my sincere gratitude to LaunchED Global for providing me with the opportunity to undertake this Data Analytics Internship and complete this Capstone Project.

This project has helped me strengthen my understanding of data analytics concepts, including data preprocessing, exploratory data analysis, data visualization, and sentiment analysis using Natural Language Processing (NLP). It provided valuable practical exposure to applying analytical techniques to real-world customer review data.

I would also like to thank my mentors, instructors, and everyone who supported and guided me throughout this internship. Their encouragement and feedback played an important role in the successful completion of this project.

Finally, I would like to express my gratitude to my college for encouraging experiential learning and providing an environment that supports the development of technical and analytical skills.

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1: ABOUT THE PROJECT

1.1 Introduction

The rapid growth of e-commerce platforms has significantly increased the importance of customer reviews in influencing purchasing decisions. Online reviews provide valuable insights into customer experiences, product quality, and overall satisfaction. Businesses can utilize these reviews to understand customer expectations, identify recurring issues, and improve their products and services.

Flipkart, one of India's leading e-commerce platforms, receives thousands of customer reviews every day across a wide variety of product categories. Analyzing this large volume of textual data manually is both time-consuming and inefficient. Data analytics and Natural Language Processing (NLP) techniques make it possible to automatically process customer reviews and extract meaningful information.

This project performs sentiment analysis on Flipkart product reviews by combining exploratory data analysis with text-based sentiment classification. The study explores customer rating patterns, analyzes review characteristics, and classifies review sentiments using the TextBlob library. The resulting insights demonstrate how customer feedback can support better business decisions and improve customer experience.

1.2 Problem Statement

Customer reviews contain valuable information regarding product quality, customer satisfaction, and user experience. However, manually analyzing thousands of customer reviews is inefficient and prone to inconsistency.

The primary problem addressed in this project is the automatic identification of customer sentiment from textual reviews and the analysis of its relationship with customer ratings. The project aims to transform unstructured customer feedback into meaningful insights that can support business decision-making and enhance customer satisfaction.

1.3 Objectives

The primary objectives of this project are:

- To understand the Flipkart customer review dataset.
- To perform data cleaning and preprocessing.
- To conduct Exploratory Data Analysis (EDA).
- To visualize customer rating patterns.
- To preprocess review text for sentiment analysis.
- To classify customer reviews into Positive, Neutral, and Negative sentiments using TextBlob.
- To compare textual sentiment with customer ratings.
- To generate meaningful business insights and recommendations.

1.4 Scope of the Project

This project focuses on analyzing customer reviews obtained from the Flipkart product review dataset. The scope includes data cleaning, exploratory data analysis, text preprocessing, sentiment analysis, and visualization of customer feedback.

The project is limited to the available dataset and does not utilize real-time customer reviews. Nevertheless, the methodology presented can be extended to larger datasets or integrated with live e-commerce platforms for continuous customer sentiment monitoring.

Excellent. This chapter is where many students write only a paragraph or two. We're going to make yours informative, well-organized, and directly based on your project.

2: DATASET DESCRIPTION

2.1 Dataset Overview

The dataset used in this project consists of customer reviews collected from products listed on the Flipkart e-commerce platform. These reviews represent customer opinions and experiences after purchasing various products. Each review is accompanied by a numerical rating, making the dataset suitable for both exploratory data analysis and sentiment analysis.

Customer review data is an important source of business intelligence because it provides direct feedback about product quality, customer satisfaction, delivery experience, and overall user perception. By analyzing this data, organizations can identify trends, improve products, and make informed business decisions.

The dataset used for this project contains 2,304 customer reviews and includes both textual and numerical information, allowing comprehensive analysis of customer feedback.

2.2 Dataset Features

Column Name	Description	Data Type
Product_name	Name and description of the product reviewed by the customer.	Object
Review	Customer's written review describing their experience with the product.	Object
Rating	Numerical rating assigned by the customer (1–5 stars).	Integer
(Unnamed: 0)	Index column generated during dataset creation. Removed during preprocessing.	Integer

2.3 Dataset Statistics

After preprocessing:

Attribute	Value
Total Records	2,304
Total Features	4 (before cleaning)
Missing Values	None
Duplicate Records	None (after verification)
Target for Analysis	Customer Review Text
Rating Scale	1–5 Stars

During the initial exploration, the dataset was examined for missing values, duplicate records, and inconsistent data. The analysis showed that there were no missing values requiring imputation. Duplicate records were also checked and handled as

part of the data-cleaning process. The unnecessary index column (Unnamed: 0) was removed before further analysis to improve dataset quality.

2.4 Tools and Technologies Used

Tool / Technology	Purpose
Google Colab	Development environment for executing the project.
Python	Programming language used for data analysis.
Pandas	Data loading, cleaning, and manipulation.
NumPy	Numerical computations and data processing.
Matplotlib	Data visualization.
Seaborn	Statistical data visualization.
TextBlob	Sentiment analysis using text polarity.
WordCloud	Visualization of frequently occurring words.

2.5 Libraries Used

The following Python libraries were utilized during the development of this project:

Pandas

Pandas was used for importing the dataset, handling tabular data, performing preprocessing operations, and manipulating data structures efficiently.

NumPy

NumPy supported numerical computations and efficient handling of arrays during data analysis.

Matplotlib

Matplotlib was used to generate various visualizations such as bar charts, pie charts, histograms, and box plots.

Seaborn

Seaborn was employed to create aesthetically appealing statistical visualizations, including count plots, bar charts, and heatmaps.

TextBlob

TextBlob is a Natural Language Processing (NLP) library used to calculate the polarity score of customer reviews. Based on the polarity values, each review was classified as Positive, Neutral, or Negative.

WordCloud

The WordCloud library was used to visually represent the most frequently occurring words in positive and negative customer reviews, making it easier to identify common themes.

2.6 Project Workflow

Flipkart Review Dataset

Data Understanding

Data Cleaning



Exploratory Data Analysis



Text Preprocessing (NLP)



Sentiment Analysis (TextBlob)



Data Visualization



Business Insights & Conclusion

3: METHODOLOGY

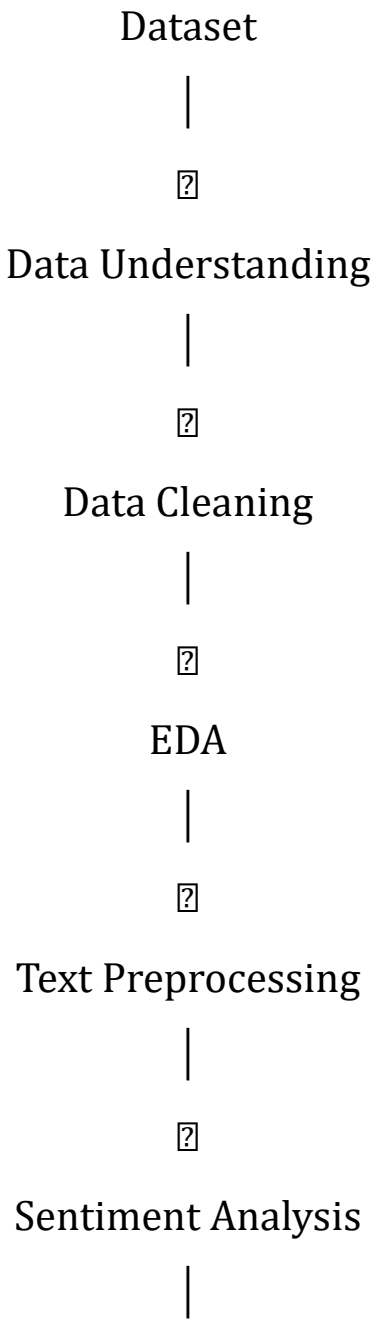
3.1 Introduction

Methodology refers to the systematic approach followed to accomplish the objectives of the project. In this study, a structured data analytics workflow was adopted, beginning with data collection and preprocessing, followed by exploratory data analysis, text preprocessing, sentiment analysis, visualization, and interpretation of the results.

The methodology ensured that the customer review data was transformed into meaningful information through a series of analytical steps. Each stage contributed to improving the quality of the analysis and generating actionable business insights.

3.2 Methodology Workflow

Figure 3.1: Overall Project Workflow

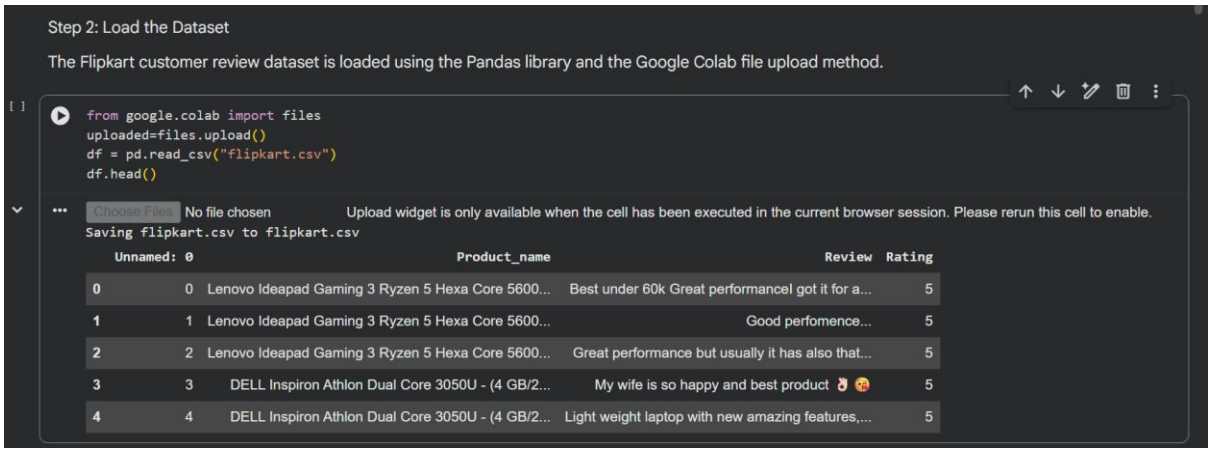


Visualization

Business Insights

3.3 Data Collection

The dataset used in this project was provided as part of the Data Analytics Capstone Project. It contains customer reviews collected from products listed on Flipkart. The dataset includes product names, customer review text, and numerical ratings, making it suitable for sentiment analysis using Natural Language Processing (NLP) techniques.



- Dataset in Google Colab
- OR df.head()

Figure 3.2: Sample of the Flipkart Review Dataset

3.4 Data Understanding

Before beginning the analysis, the dataset was examined to understand its structure, dimensions, data types, and overall quality. Functions such as head(), shape, columns, info(), and describe() were used to gain an overview of the data.

- df.info()

```
print("\nDataset Information:")
df.info()

Rows and Columns: (2304, 4)

Columns:
Index(['Unnamed: 0', 'Product_name', 'Review', 'Rating'], dtype='object')

Data Types:
Unnamed: 0       int64
Product_name    object
Review          object
Rating           int64
dtype: object

Dataset Information:
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 2304 entries, 0 to 2303
Data columns (total 4 columns):
 #   Column        Non-Null Count  Dtype
---  ------        --------------  -----
 0   Unnamed: 0    2304 non-null   int64
 1   Product_name  2304 non-null   object
 2   Review        2304 non-null   object
 3   Rating        2304 non-null   int64
dtypes: int64(2), object(2)
memory usage: 72.1+ KB
```

- df.describe()

Step 5: Descriptive Statistics

The statistical summary provides information such as count, mean, minimum, maximum, and quartiles for numerical columns.

```
df.describe()
```

	Unnamed: 0	Rating
count	2304.000000	2304.000000
mean	1151.500000	4.259549
std	665.251832	1.180017
min	0.000000	1.000000
25%	575.750000	4.000000
50%	1151.500000	5.000000
75%	1727.250000	5.000000
max	2303.000000	5.000000

Figure 3.3: Dataset Information

Figure 3.4: Statistical Summary

3.5 Data Cleaning

This is one of the strongest sections because you actually performed all these steps.

Data cleaning is an essential stage in data analytics, as inaccurate or inconsistent data can affect the quality of the analysis. Several preprocessing operations were performed to prepare the dataset for further analysis.

Cleaning Steps Performed:

- Checked missing values
- Verified duplicate records
- Removed duplicate entries (if present)
- Removed unnecessary index column
- Verified data types
- Ensured dataset consistency

- Missing values output

Step 6: Missing Value Analysis

Missing values can affect the quality of analysis. Therefore, each column is checked for null values.

```
missing_values = df.isnull().sum()

print("Missing Values:")
print(missing_values)
```

```
Missing Values:
Unnamed: 0      0
Product_name    0
Review          0
Rating          0
dtype: int64
```

- Duplicate check

Step 8: Duplicate Record Analysis

Duplicate records can lead to biased analysis. Hence, duplicate rows are identified and removed if necessary.

```
duplicates = df.duplicated().sum()

print("Duplicate Records:", duplicates)
```

```
Duplicate Records: 0
```

```
df = df.drop_duplicates()

print("Dataset Shape after removing duplicates:")
print(df.shape)
```

```
Dataset Shape after removing duplicates:
(2304, 4)
```

- Column removal

Step 10: Removing Unnecessary Columns

The column 'Unnamed: 0' acts only as an index and does not contribute to the analysis. Therefore, it is removed.

```
df.drop(columns=['Unnamed: 0'], inplace=True)
df.columns
```

```
Index(['Product_name', 'Review', 'Rating'], dtype='object')
```

Figure 3.5: Missing Value Analysis

Figure 3.6: Duplicate Record Check

3.6 Exploratory Data Analysis

Exploratory Data Analysis (EDA) was performed to understand customer rating patterns, review characteristics, and product popularity. Various statistical summaries and graphical visualizations were created to identify meaningful trends and relationships within the dataset.

The following visualizations were generated:

- Rating Distribution
- Rating Percentage
- Review Length Distribution

- Review Length vs Rating
- Top Reviewed Products
- Correlation Heatmap

Each visualization was interpreted to understand customer behaviour and identify patterns that could support business decision-making.

3.7 Text Preprocessing

Since customer reviews consist of unstructured textual data, preprocessing was necessary before performing sentiment analysis. The review text was standardized by converting all characters to lowercase, removing punctuation marks, and eliminating unnecessary spaces. These preprocessing steps improve consistency and enhance the quality of sentiment analysis.

The preprocessing pipeline consisted of:

- Converting text to lowercase
- Removing punctuation
- Removing extra spaces
- Creating the cleaned review column

- Before cleaning

TEXT PREPROCESSING

Text Preprocessing

Customer reviews often contain punctuation, capital letters, extra spaces, and other unnecessary characters that can affect sentiment analysis.

Text preprocessing helps improve the quality of analysis by cleaning and standardizing the review text.

```
# Convert reviews to lowercase
df['Clean_Review'] = df['Review'].str.lower()

df[['Review','Clean_Review']].head()
```

	Review	Clean_Review
0	Best under 60k Great performancel got it for a...	best under 60k great performancei got it for a...
1	Good perfomence...	good perfomence...
2	Great performance but usually it has also that...	great performance but usually it has also that...
3	My wife is so happy and best product 👌😍	my wife is so happy and best product 👌😍
4	Light weight laptop with new amazing features,...	light weight laptop with new amazing features,...

- After cleaning

CHECK CLEANED REVIEWS

```
df[['Review','Clean_Review']].head(10)
```

	Review	Clean_Review
0	Best under 60k Great performancel got it for a...	best under 60k great performancei got it for a...
1	Good perfomence...	good perfomence
2	Great performance but usually it has also that...	great performance but usually it has also that...
3	My wife is so happy and best product 👌😍	my wife is so happy and best product 👌😍
4	Light weight laptop with new amazing features,...	light weight laptop with new amazing features ...
5	Amazing laptop, am so much happy, thanks for F...	amazing laptop am so much happy thanks for fli...
6	Over all a good laptop for personal use	over all a good laptop for personal use
7	Thank you so much Flipkart	thank you so much flipkart
8	Amazing product	amazing product
9	Good for normal work , students, online classe...	good for normal work students online classes w...

Figure 3.7: Review Text Before and After Preprocessing

3.8 Sentiment Analysis

This is probably the best section in your report.

Sentiment Analysis was performed using the TextBlob library, which applies Natural Language Processing techniques to

determine the polarity of textual data. Each customer review was assigned a polarity score ranging from -1 to +1, where negative values indicate negative sentiment, positive values indicate positive sentiment, and values close to zero indicate neutral sentiment.

Based on the polarity score, customer reviews were classified into three categories:

Polarity	Sentiment
< 0	Negative
$= 0$	Neutral
> 0	Positive

The classified sentiments were then compared with customer ratings to evaluate whether textual opinions aligned with numerical ratings.

- Polarity output

CALCULATE POLARITY

```
[ ]
from textblob import TextBlob

def get_polarity(review):
    return TextBlob(review).sentiment.polarity

df['Polarity'] = df['Clean_Review'].apply(get_polarity)

df[['Clean_Review','Polarity']].head()
```

	Clean_Review	Polarity
0	best under 60k great performancei got it for a...	0.438788
1	good perfomence	0.700000
2	great performance but usually it has also that...	0.183333
3	my wife is so happy and best product 👌😘	0.900000
4	light weight laptop with new amazing features ...	0.534091

- Sentiment output

Each review has been classified as Positive, Neutral, or Negative based on its sentiment polarity score.

```
[ ]
def get_sentiment(score):

    if score > 0:
        return "Positive"

    elif score < 0:
        return "Negative"

    else:
        return "Neutral"

df['Sentiment'] = df['Polarity'].apply(get_sentiment)

df[['Review','Sentiment']].head()
```

	Review	Sentiment
0	Best under 60k Great performancel got it for a...	Positive
1	Good perfomence...	Positive
2	Great performance but usually it has also that...	Positive
3	My wife is so happy and best product 👌😘	Positive
4	Light weight laptop with new amazing features,...	Positive

Figure 3.8: Sentiment Classification

4: EXPLORATORY DATA ANALYSIS (EDA)

4.1 Introduction

Exploratory Data Analysis (EDA) is an essential phase of any data analytics project. It involves summarizing, visualizing, and interpreting the dataset to uncover patterns, trends, and relationships. Through EDA, analysts gain a deeper understanding of the data before applying advanced analytical techniques.

In this project, EDA was performed to examine customer rating patterns, review characteristics, and product popularity. Various statistical visualizations were created to identify customer behaviour and derive meaningful business insights.

4.2 Distribution of Customer Ratings

Graph 1: *Rating Distribution*

II) EXPLORATORY DATA ANALYSIS (EDA)

RATING DISTRIBUTION

Distribution of Customer Ratings

This visualization shows how customers have rated the products on a scale of 1 to 5 stars.

[ ]

```
plt.figure(figsize=(8,5))

sns.countplot(
    x='Rating',
    data=df,
    palette='viridis'
)

plt.title("Distribution of Customer Ratings", fontsize=14)
plt.xlabel("Rating")
plt.ylabel("Number of Reviews")

plt.show()
```

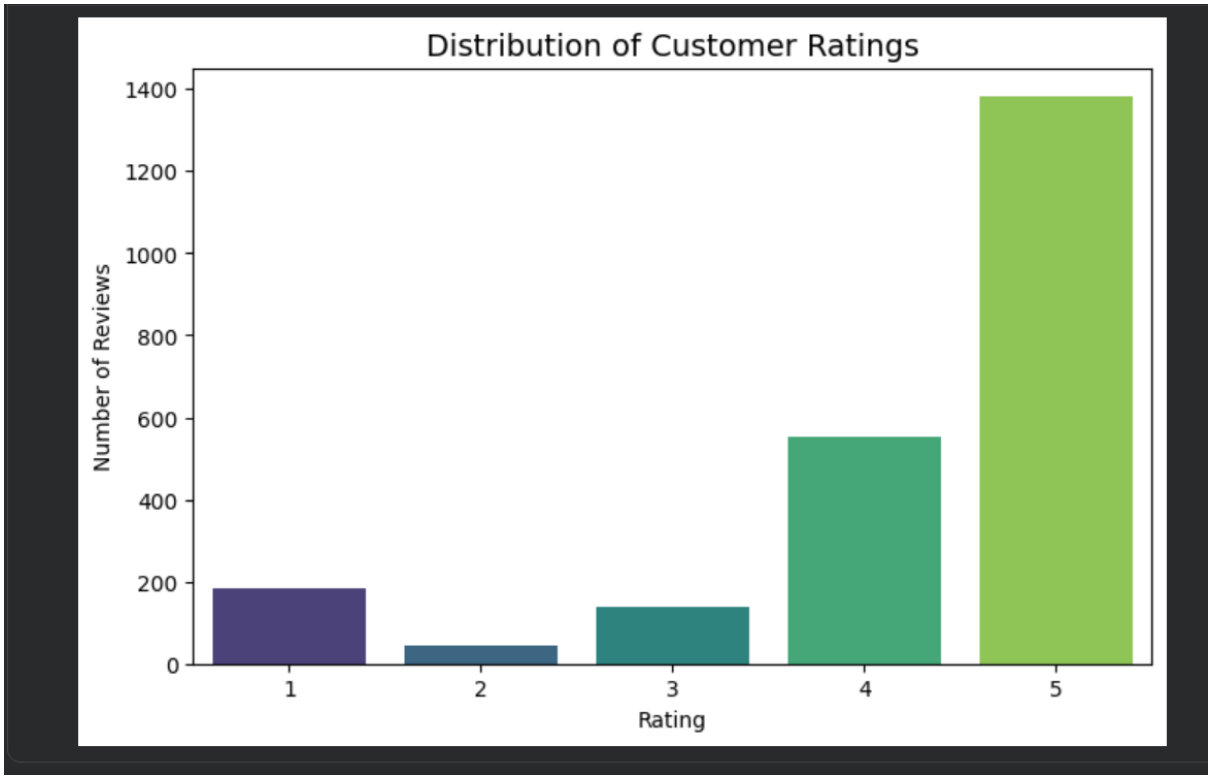


Figure 4.1: Distribution of Customer Ratings

Description

The bar chart illustrates the frequency of customer ratings on a scale of one to five stars. It provides an overview of customer satisfaction and highlights the distribution of ratings across all products in the dataset.

Interpretation

The analysis indicates that higher ratings occur more frequently than lower ratings, suggesting that most customers were satisfied with their purchases. Ratings of four and five stars dominate the dataset, while one-star and two-star ratings are comparatively less common. This pattern reflects an overall positive customer experience.

4.3 Percentage Distribution of Ratings

Pie Chart:

PERCENTAGE DISTRIBUTION OF RATINGS

- The pie chart provides a percentage-wise representation of customer ratings.
- It helps understand the proportion of positive, neutral, and negative ratings in the dataset.

```
[ ]
rating_counts = df['Rating'].value_counts()

plt.figure(figsize=(7,7))

plt.pie(
    rating_counts,
    labels=rating_counts.index,
    autopct='%1.1f%%',
    startangle=90
)

plt.title("Percentage Distribution of Ratings")

plt.show()
```

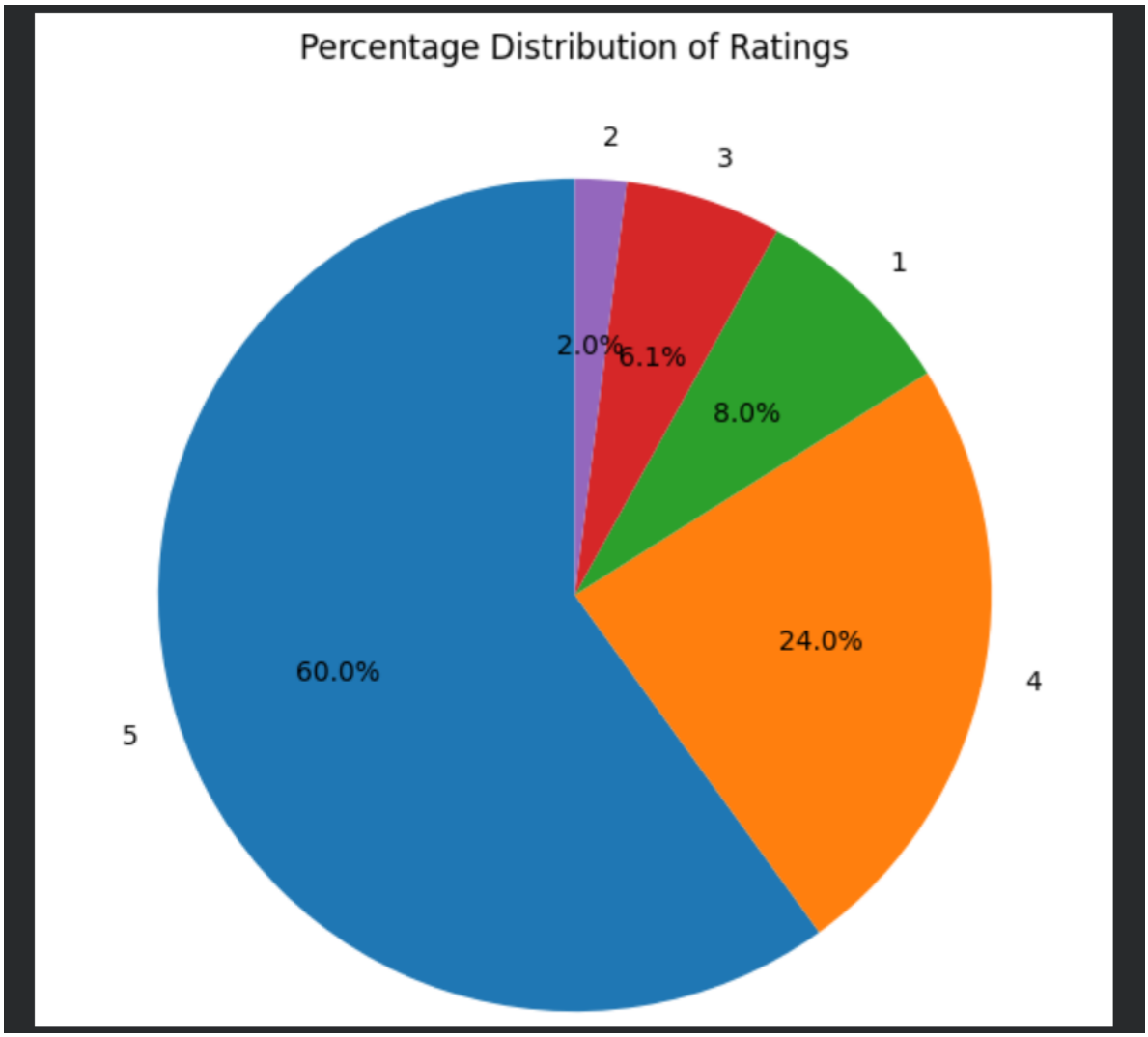


Figure 4.2: Percentage Distribution of Customer Ratings

Description

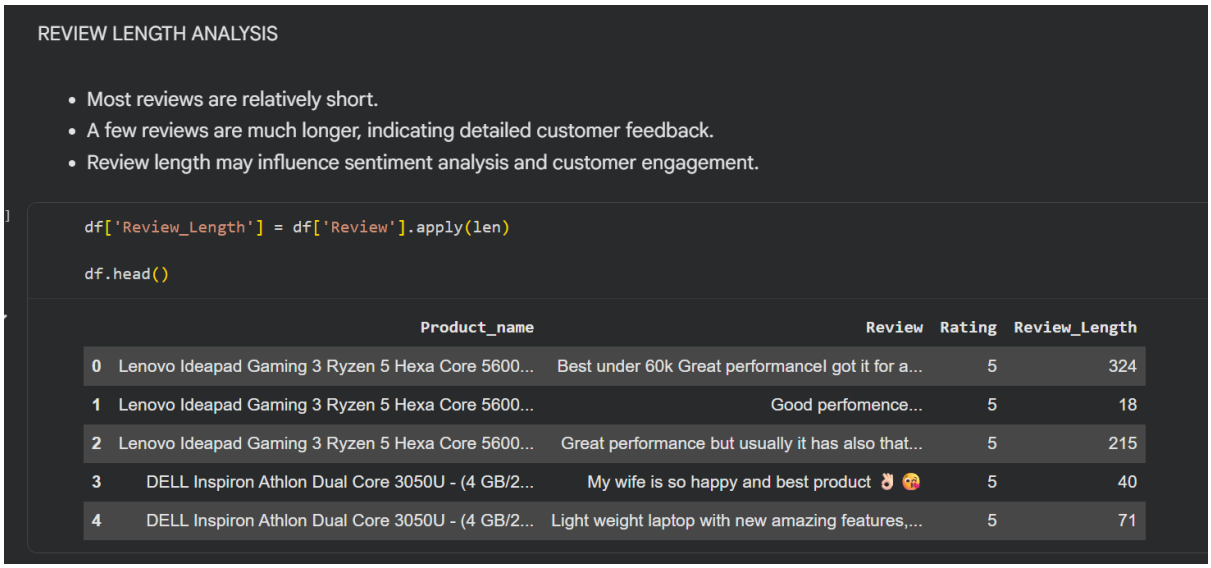
The pie chart represents the percentage contribution of each rating category, providing a proportional view of customer satisfaction.

Interpretation

The majority of reviews belong to the higher rating categories, confirming the positive trend observed in the previous analysis. Lower-rated reviews form only a small proportion of the dataset, indicating fewer dissatisfied customers.

4.4 Review Length Analysis

Histogram:


REVIEW LENGTH ANALYSIS

- Most reviews are relatively short.
- A few reviews are much longer, indicating detailed customer feedback.
- Review length may influence sentiment analysis and customer engagement.

```
df['Review_Length'] = df['Review'].apply(len)

df.head()
```

	Product_name	Review	Rating	Review_Length
0	Lenovo Ideapad Gaming 3 Ryzen 5 Hexa Core 5600...	Best under 60k Great performancel got it for a...	5	324
1	Lenovo Ideapad Gaming 3 Ryzen 5 Hexa Core 5600...	Good perfomence...	5	18
2	Lenovo Ideapad Gaming 3 Ryzen 5 Hexa Core 5600...	Great performance but usually it has also that...	5	215
3	DELL Inspiron Athlon Dual Core 3050U - (4 GB/2...	My wife is so happy and best product 👌😍	5	40
4	DELL Inspiron Athlon Dual Core 3050U - (4 GB/2...	Light weight laptop with new amazing features,...	5	71

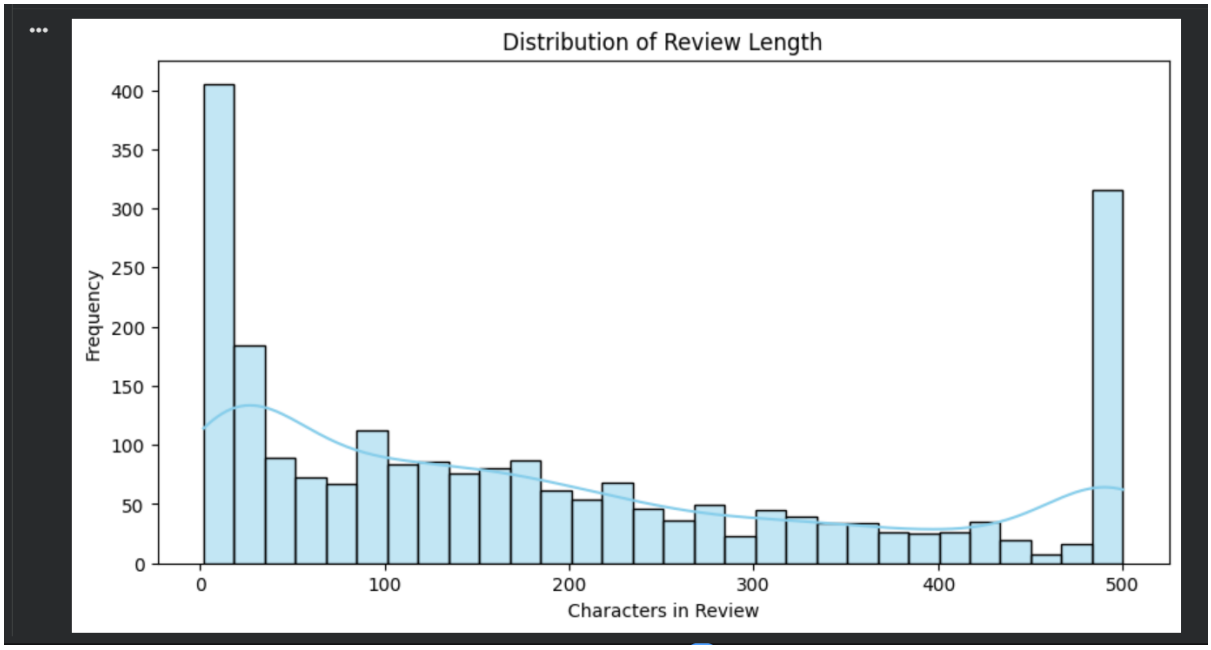


Figure 4.3: Distribution of Review Length

Description

Review length was calculated based on the number of characters in each customer review. The histogram illustrates the distribution of review lengths across the dataset.

Interpretation

Most customers provided concise reviews, while a smaller number wrote detailed feedback. Longer reviews often contain more comprehensive opinions, making them particularly useful for qualitative sentiment analysis.

4.5 Review Length by Customer Rating

Box Plot:

REVIEW LENGTH VS RATING

- The boxplot compares review lengths across different ratings.
- Customers giving lower ratings may write longer reviews to explain their dissatisfaction.
- Higher ratings often correspond to shorter appreciation comments.

```
[ ]
    plt.figure(figsize=(10,6))

    sns.boxplot(
        x='Rating',
        y='Review_Length',
        data=df,
        palette='Set2'
    )

    plt.title("Review Length by Customer Rating")

    plt.show()
```



Figure 4.4: Review Length by Rating

Description

The box plot compares review lengths for different customer ratings to determine whether dissatisfied or satisfied customers tend to write longer reviews.

Interpretation

The analysis suggests that review length varies across rating categories. Although many positive reviews are brief expressions of satisfaction, some lower-rated reviews are noticeably longer, indicating that dissatisfied customers often provide more detailed explanations of their experiences.

4.6 Top 10 Most Reviewed Products

Bar Chart:

TOP 10 MOST REVIEWED PRODUCTS

- The visualization identifies the products receiving the highest number of customer reviews.
- Products with more reviews generally have greater customer engagement.

```
[ ]
top_products = df['Product_name'].value_counts().head(10)

plt.figure(figsize=(12,6))

sns.barplot(
    x=top_products.values,
    y=top_products.index,
    palette='magma'
)

plt.title("Top 10 Most Reviewed Products")

plt.xlabel("Number of Reviews")
plt.ylabel("Product Name")

plt.show()
```

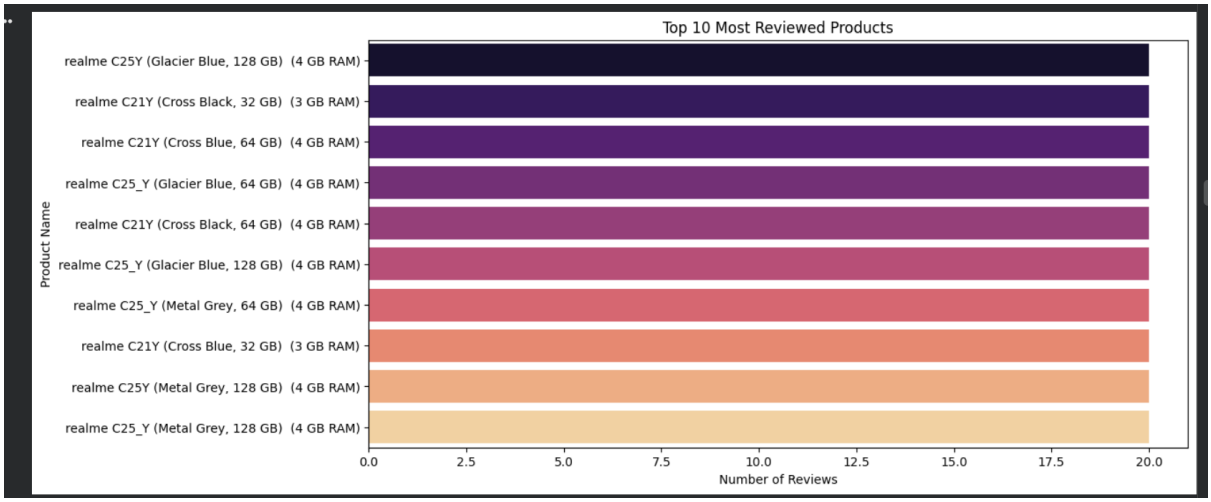


Figure 4.5: Top 10 Most Reviewed Products

Description

This visualization identifies the products receiving the highest number of customer reviews. The number of reviews can be considered an indicator of customer engagement and product popularity.

Interpretation

Products with a large volume of reviews attract greater customer interaction. These products provide more reliable customer feedback due to the larger sample size and can be prioritized for further analysis by businesses.

4.7 Correlation Heatmap

Heatmap:

CORRELATION HEATMAP

- The heatmap shows the relationship between review length and customer ratings.
- A correlation value close to zero indicates a weak relationship.

```
[ ]
plt.figure(figsize=(6,4))

sns.heatmap(
    df[['Rating','Review_Length']].corr(),
    annot=True,
    cmap='Blues'
)

plt.title("Correlation Heatmap")

plt.show()
```

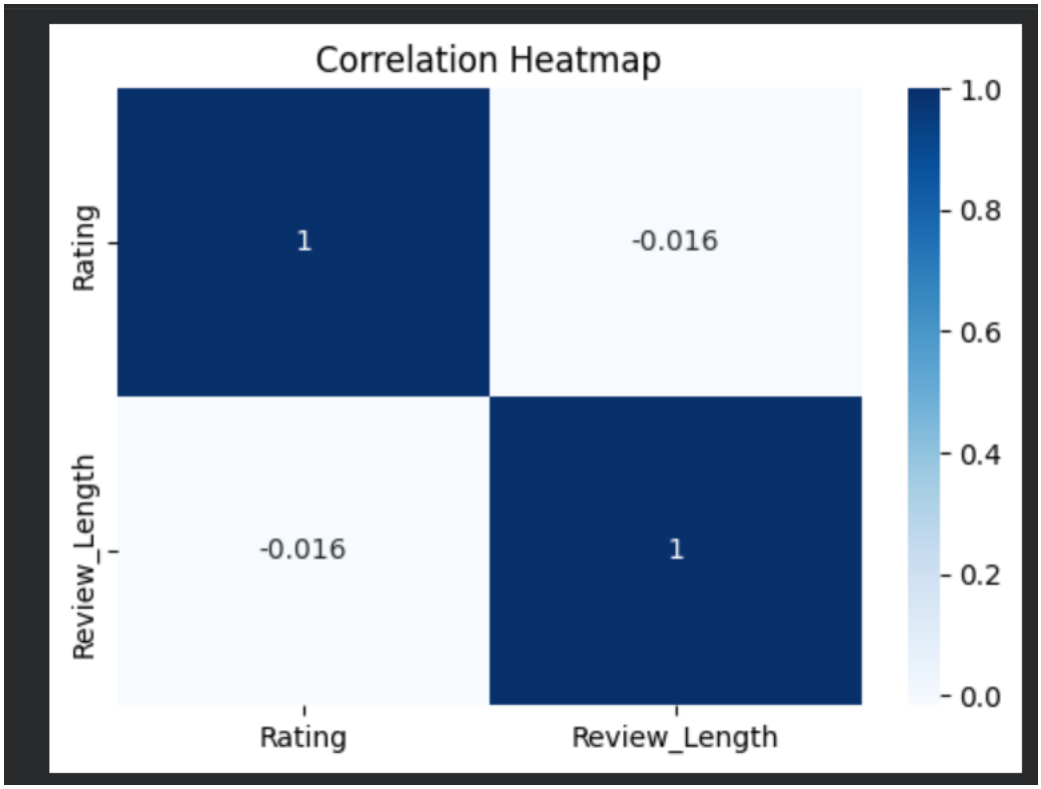


Figure 4.6: Correlation Between Numerical Features

Description

A correlation heatmap was generated to examine the relationship between numerical variables, including customer ratings and review length.

Interpretation

The analysis indicates a weak correlation between review length and customer ratings. This suggests that the length of a review alone is not a reliable indicator of whether a customer is satisfied or dissatisfied.

4.8 Summary of Exploratory Data Analysis

The exploratory data analysis provided valuable insights into customer behaviour and review patterns. Most customers expressed positive opinions through higher ratings, while only a

limited number reported negative experiences. The review length analysis demonstrated that customer feedback varies considerably in detail, and product popularity differed across the dataset. These findings established a strong foundation for the sentiment analysis performed in the next chapter.

Table 4.1: Summary of EDA Findings

Analysis	Observation
Rating Distribution	Most customers assigned 4 or 5-star ratings.
Percentage Distribution	Positive ratings formed the largest share of the dataset.
Review Length	Most reviews were short and concise.
Review Length vs Rating	Lower-rated reviews were often more descriptive.
Top Reviewed Products	Certain products attracted significantly more customer feedback.
Correlation	Review length had only a weak relationship with rating.

5: SENTIMENT ANALYSIS

5.1 Introduction

Sentiment Analysis is a Natural Language Processing (NLP) technique used to identify and classify opinions expressed in

textual data. It enables organizations to automatically determine whether customer feedback is positive, negative, or neutral. In the context of e-commerce, sentiment analysis helps businesses understand customer satisfaction, identify recurring issues, and make informed decisions based on customer opinions.

In this project, sentiment analysis was performed on Flipkart customer reviews using the TextBlob library. The polarity score generated by TextBlob was used to classify each review into Positive, Neutral, or Negative sentiment categories.

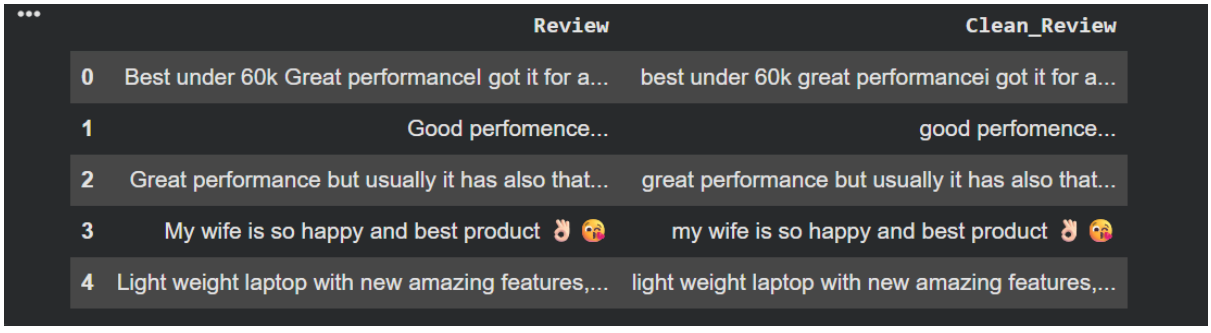
5.2 Text Preprocessing

Before performing sentiment analysis, the review text was cleaned to improve consistency and accuracy.

The preprocessing steps included:

- Conversion of all review text to lowercase.
- Removal of punctuation marks.
- Removal of unnecessary spaces.
- Creation of a cleaned review column for analysis.

Review Text Before and After Preprocessing:



	Review	Clean_Review
0	Best under 60k Great performancel got it for a...	best under 60k great performancei got it for a...
1	Good perfomence...	good perfomence...
2	Great performance but usually it has also that...	great performance but usually it has also that...
3	My wife is so happy and best product 👌😘	my wife is so happy and best product 👌😘
4	Light weight laptop with new amazing features,...	light weight laptop with new amazing features,...

```
df[['Review','Clean_Review']].head(10)
```

	Review	Clean_Review
0	Best under 60k Great performancel got it for a...	best under 60k great performancei got it for a...
1	Good perfomence...	good perfomence
2	Great performance but usually it has also that...	great performance but usually it has also that...
3	My wife is so happy and best product 👌😘	my wife is so happy and best product 👌😘
4	Light weight laptop with new amazing features,...	light weight laptop with new amazing features ...
5	Amazing laptop, am so much happy, thanks for F...	amazing laptop am so much happy thanks for fli...
6	Over all a good laptop for personal use	over all a good laptop for personal use
7	Thank you so much Flipkart	thank you so much flipkart
8	Amazing product	amazing product
9	Good for normal work , students, online classe...	good for normal work students online classes w...

Explanation

Text preprocessing standardizes the textual data and reduces inconsistencies caused by variations in capitalization, punctuation, and formatting. These steps improve the quality of sentiment analysis by ensuring that the text is processed in a uniform manner.

5.3 Sentiment Classification using TextBlob

The cleaned review text was analyzed using the TextBlob library.

TextBlob assigns each review a polarity score ranging from -1 to +1.

Polarity Score	Sentiment
Less than 0	Negative
Equal to 0	Neutral
Greater than 0	Positive

Based on the calculated polarity score, every customer review was automatically classified into one of the three sentiment categories.

Sample Sentiment Classification Output:

	Review	Sentiment
0	Best under 60k Great performancel got it for a...	Positive
1	Good perfomence...	Positive
2	Great performance but usually it has also that...	Positive
3	My wife is so happy and best product 👌 😘	Positive
4	Light weight laptop with new amazing features,...	Positive

5.4 Distribution of Customer Sentiments

Count Plot:

```
plt.figure(figsize=(8,5))

sns.countplot(
    x='Sentiment',
    data=df,
    palette='Set2'
)

plt.title("Distribution of Customer Sentiments")

plt.xlabel("Sentiment")
plt.ylabel("Number of Reviews")

plt.show()
```

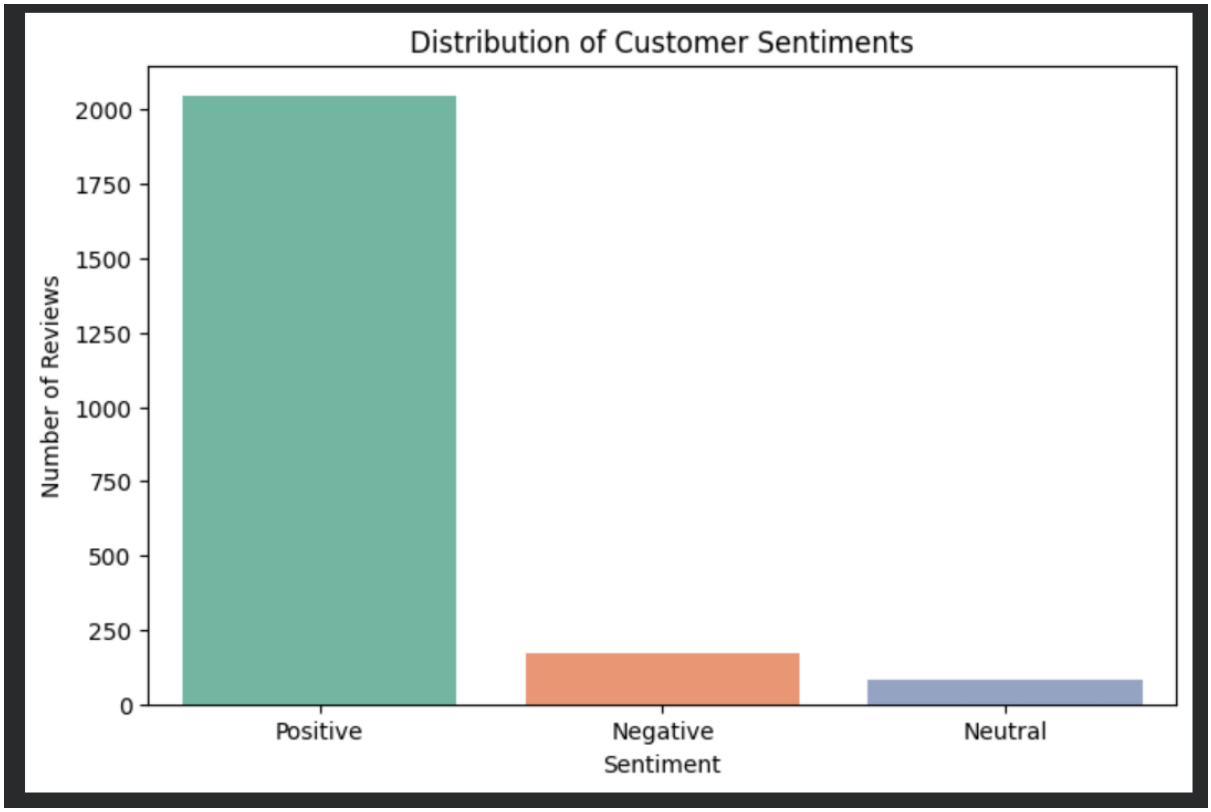


Figure 5.3: Distribution of Customer Sentiments

Description

This visualization shows the number of customer reviews classified as Positive, Neutral, and Negative based on the TextBlob polarity scores.

Interpretation

The sentiment distribution provides an overview of customer opinions expressed in textual reviews. A higher proportion of positive reviews indicates overall customer satisfaction, whereas a larger number of negative reviews would suggest recurring customer concerns.

5.5 Percentage Distribution of Sentiments

Pie Chart:

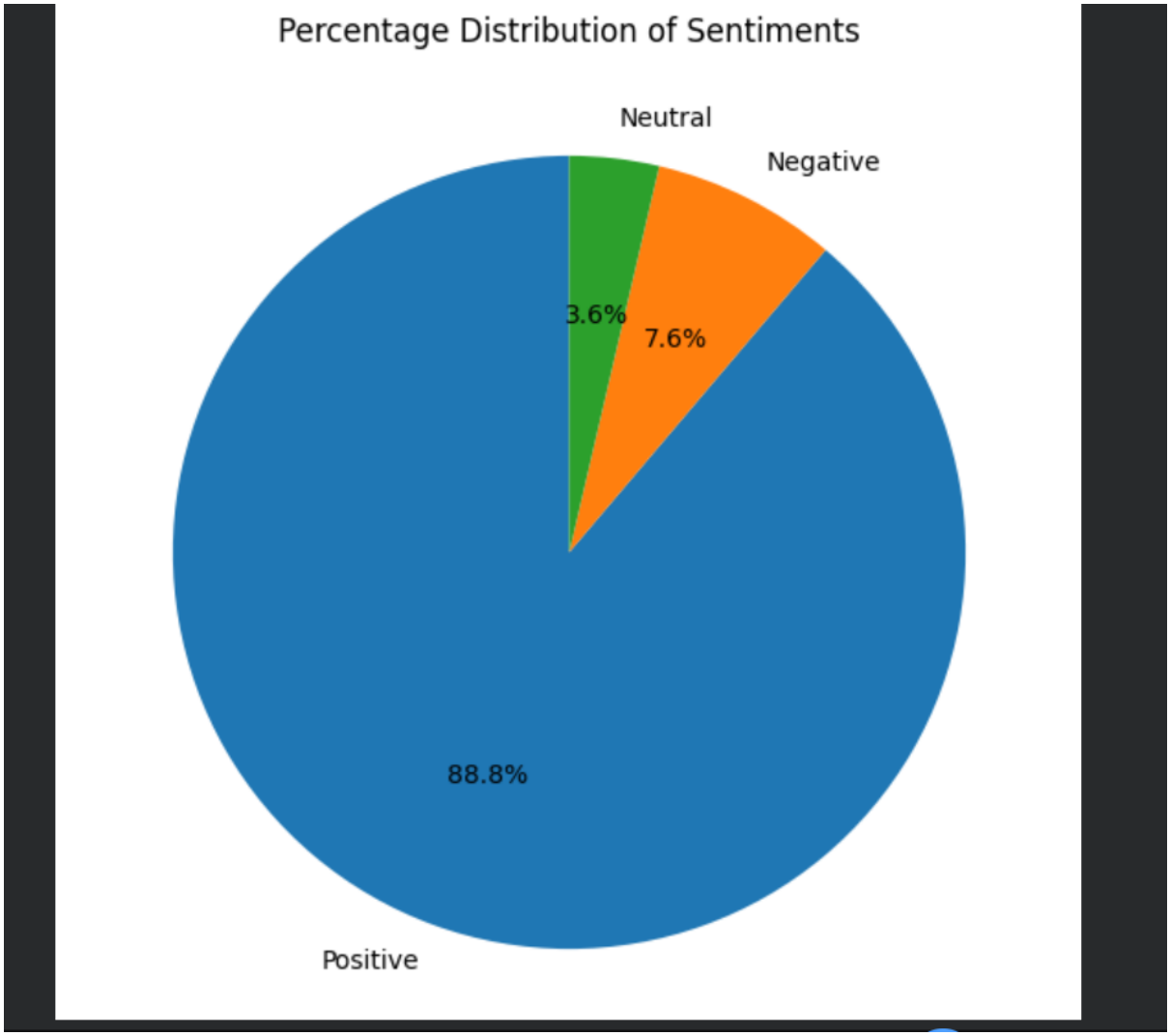


Figure 5.4: Percentage Distribution of Customer Sentiments

Description

The pie chart illustrates the proportion of positive, neutral, and negative reviews in the dataset.

Interpretation

The percentage distribution highlights the overall sentiment trend within the customer reviews. It enables businesses to quickly assess customer satisfaction levels and identify the prevalence of different sentiment categories.

5.6 Comparison of Customer Ratings and Sentiments

Rating vs Sentiment Graph:

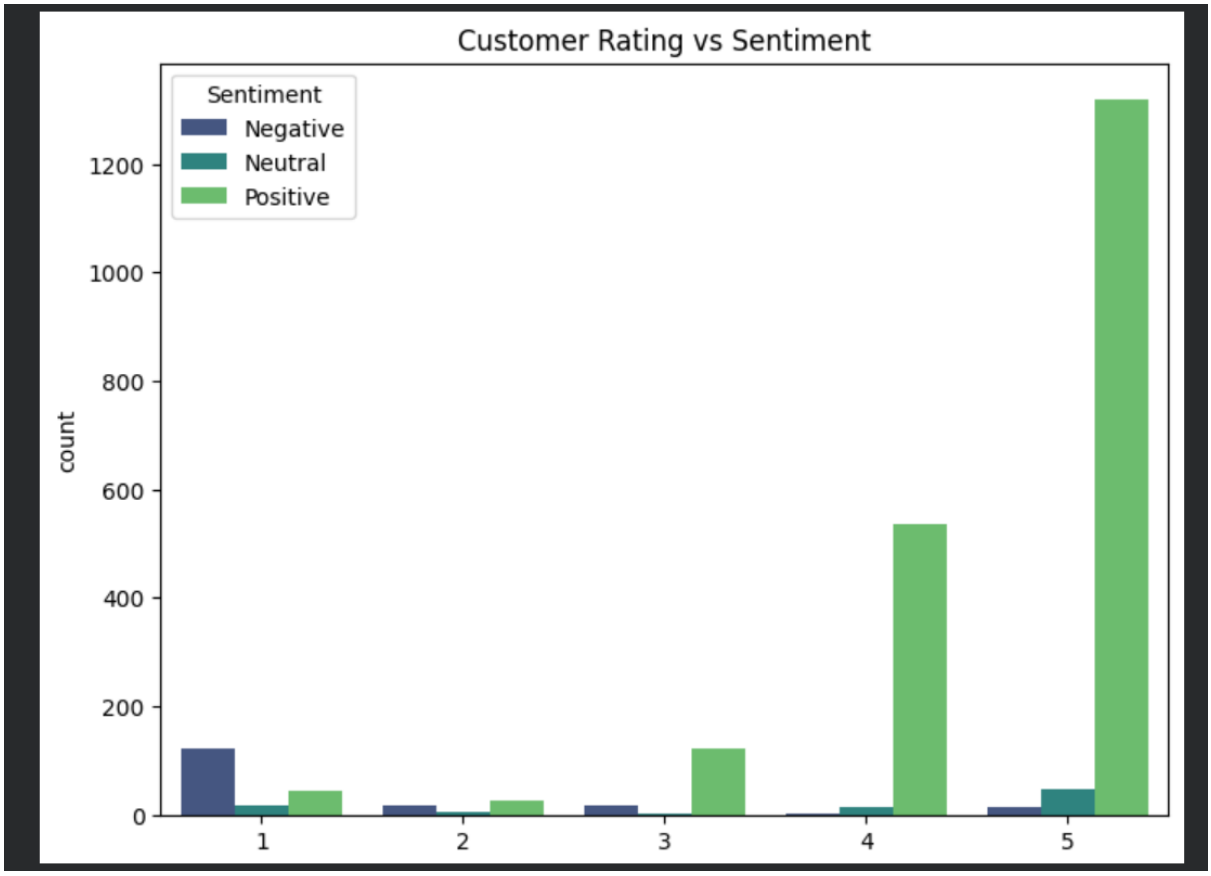


Figure 5.5: Comparison of Ratings and Sentiments

Description

This visualization compares numerical customer ratings with the sentiments extracted from the corresponding review text.

Interpretation

In most cases, higher customer ratings correspond to positive textual sentiment, while lower ratings align with negative sentiment. However, a few differences may occur because customers sometimes assign high ratings while expressing minor concerns in the review text, or vice versa.

5.7 Heatmap of Ratings and Sentiments

Heatmap:

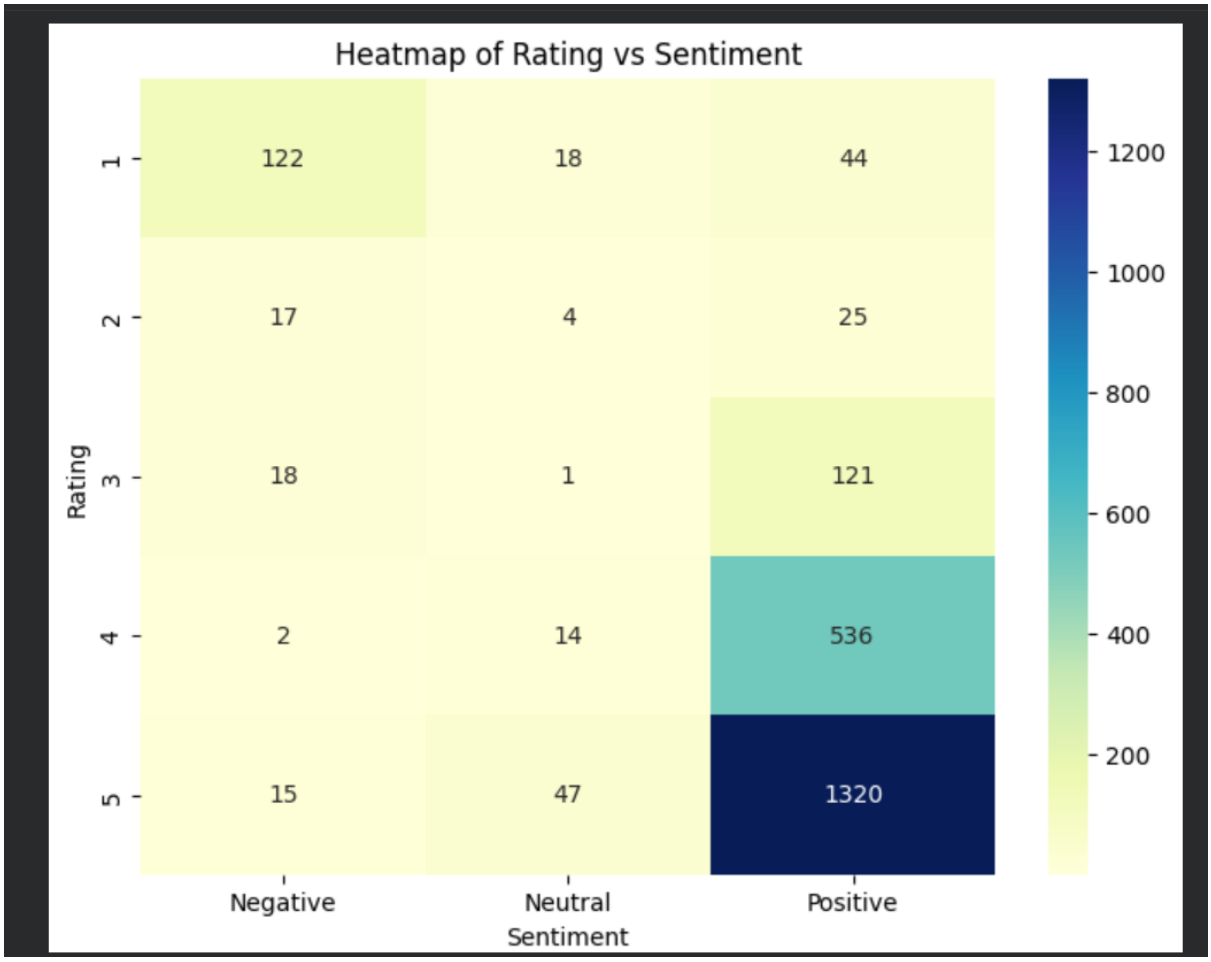


Figure 5.6: Heatmap of Rating vs Sentiment

Description

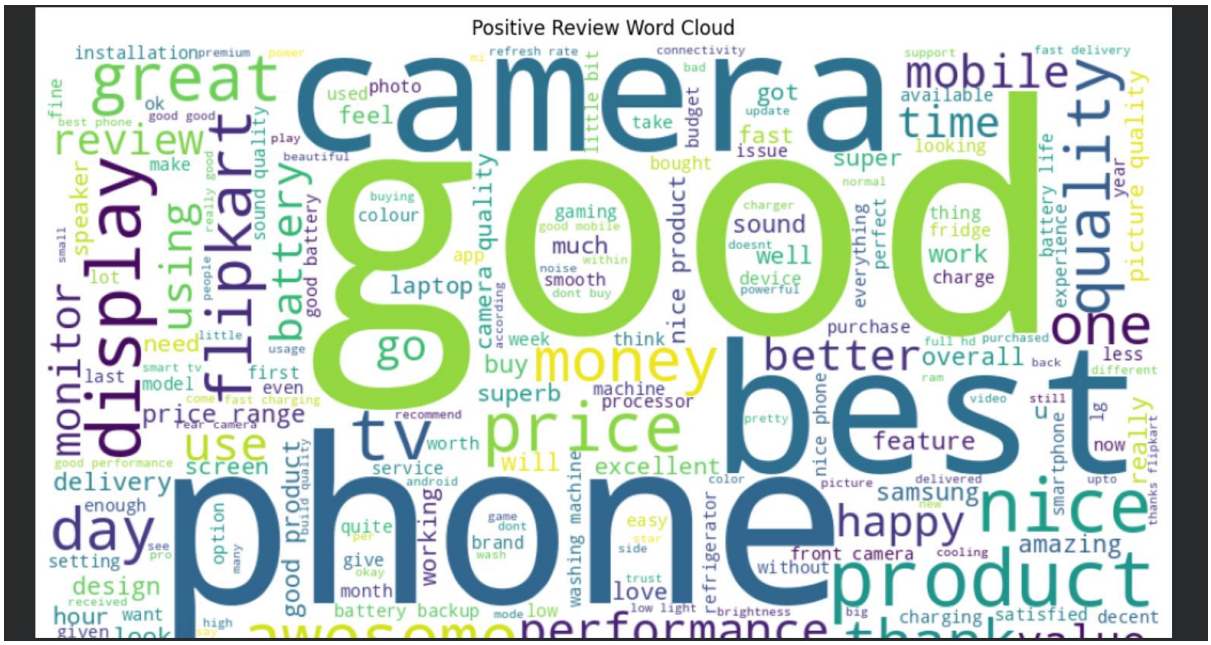
The heatmap presents the frequency of each combination of customer rating and sentiment classification.

Interpretation

The visualization clearly shows the relationship between ratings and textual sentiment. Strong agreement between high ratings and positive sentiment indicates consistency in customer feedback, while mismatches provide useful insights into customer behaviour.

5.8 Positive Word Cloud

Positive Word Cloud:


Positive Review Word Cloud
camera
good
best
phone
product
nice
mobile
great
flipkart
display
battery
quality
price
money
tv
review
time
one
better
awesome
performance
thankyou
value
feature
happy
amazing
delivery
day
use
screen
go
buy
laptop
camera
sound
work
charge
fridge
smartphone
samsung
front camera
charging
satisfied
decent
overall
purchase
processor
machine
superb
excellent
working
washing machine
refrigerator
battery backup
picture quality
speaker
monitor
using
need
model
price range
good product
build quality
smart tv
fast charging
rear camera
good performance
installation
premium
refresh rate
connectivity
fast delivery
support
available
looking
normal
charger
doesnt
perfect
everything
thing
experience
battery life
year
picture
quality
sound
colour
buying
gaming
good mobile
smooth
dont buy
noise
within
nice product
bought
issue
super
fast
got
update
budget
bad
little bit
take
feel
photo
used
ok
good good
best phone
fine
make
play
beautiful
good battery
usage
little
last
first
even
recommend
service
worth
will
week
think
full hd
purchased
back
ram
video
still
lg
less
different
upto
delivered
now
really
thanks flipkart
nice phone
pretty
color
picture
android
game
dont
brand
wash
quite
per
give
okay
month
easy
star
side
trust
love
low light
brightness
big
mode
low
battery backup
without
cooling
new
option
many
setting
see
pro
design
received
hour
want
given
look
high
enough
come
people
really
good
small
lot
outside
camera
according
app
laptop
much
smooth
dont buy
noise
within
sound
doesnt
well
device
powerful
charger
thing
fridge
work
charge
battery life
experience
year
picture quality
purchase
full hd
purchased
back
less
different
ram
video
still
lg
now
upto
delivered
picture
color
nice phone
pretty
feature
u
really
thanks flipkart
samsung
happy
amazing
front camera
cooling
without
charging
satisfied
decent
refrigerator
brightness
big
low light
trust
love
easy
side
star
quite
per
give
okay
month
brand
wash
game
dont
android
service
worth
will
excellent
machine
processor
superb
buy
go
camera
quality
accordingly
week
think
nice product
bought
issue
super
fast
got
update
budget
bad
little bit
take
feel
photo
used
ok
good good
best phone
fine
make
play
beautiful
good battery
usage
little
last
first
even
recommend

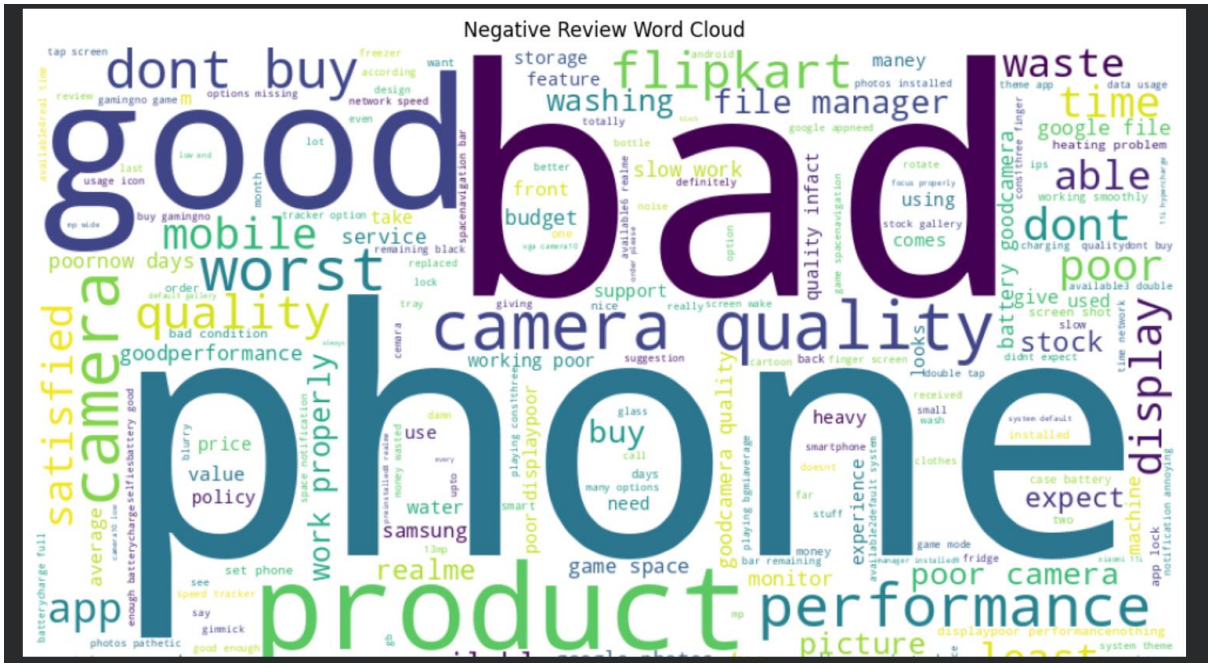


Figure 5.8: Frequently Occurring Words in Negative Reviews

Description

The negative word cloud highlights commonly occurring words in reviews classified as negative.

Interpretation

Frequently occurring negative words help identify customer complaints and recurring issues. Businesses can use these insights to improve product quality, customer support, and logistics.

5.10 Summary of Sentiment Analysis

The sentiment analysis revealed the overall emotional tone expressed in customer reviews. Using TextBlob, reviews were successfully categorized into Positive, Neutral, and Negative classes. The comparison between customer ratings and textual sentiment showed that both generally aligned, indicating consistency between numerical ratings and written feedback. The

word cloud analysis further highlighted commonly used positive and negative terms, providing valuable insights into customer experiences.

Table 5.1: Summary of Sentiment Analysis

Analysis	Observation
Text Preprocessing	Reviews were cleaned and standardized before analysis.
TextBlob Polarity	Reviews were classified into Positive, Neutral, and Negative sentiments.
Sentiment Distribution	Positive reviews formed the largest share of the dataset.
Rating vs Sentiment	Ratings generally aligned with textual sentiment.
Word Clouds	Frequently used words highlighted customer preferences and concerns.

6: RESULTS AND DISCUSSION

6.1 Introduction

This chapter presents the outcomes of the exploratory data analysis and sentiment analysis performed on the Flipkart customer review dataset. The results are interpreted to identify customer behaviour, evaluate satisfaction levels, and derive meaningful business insights. The findings demonstrate how data analytics and Natural Language Processing (NLP) techniques can transform raw customer reviews into actionable information.

6.2 Key Findings

The analysis of the Flipkart customer review dataset led to several important findings:

Finding 1: Customer Satisfaction

The rating distribution indicated that a majority of customers provided higher ratings, suggesting an overall positive customer experience. This reflects favourable customer perception of the products included in the dataset.

Finding 2: Positive Sentiment Dominance

Sentiment analysis revealed that positive reviews formed the largest category among all customer reviews. This suggests that most customers expressed satisfaction through both their ratings and written feedback.

Finding 3: Rating and Sentiment Relationship

A strong relationship was observed between customer ratings and textual sentiment. Reviews with higher ratings generally contained positive language, while lower-rated reviews tended to express dissatisfaction. However, a few mismatches highlighted that numerical ratings and written opinions are not always perfectly aligned.

Finding 4: Review Characteristics

The review length analysis showed that customers generally preferred writing concise reviews. Nevertheless, some dissatisfied customers provided more detailed explanations of their experiences, making longer reviews valuable sources of customer insights.

Finding 5: Product Engagement

Certain products received significantly more customer reviews than others. Products with higher review volumes provide more reliable customer feedback and may represent popular or widely purchased items.

6.3 Discussion of Results

The findings indicate that customer review data provides valuable information beyond numerical ratings alone. While ratings offer a quick measure of customer satisfaction, textual reviews provide detailed explanations that help businesses understand the reasons behind customer opinions.

The use of TextBlob successfully classified customer sentiments based on review text, enabling comparison between customer ratings and textual expressions. The close agreement observed between ratings and sentiments validates the effectiveness of sentiment analysis for understanding customer feedback.

Word cloud analysis further highlighted commonly used positive and negative terms, offering additional insights into customer expectations and recurring concerns. Such information can assist organizations in identifying strengths and addressing weaknesses in their products and services.

6.4 Business Insights

The analysis provides several practical insights that can support decision-making in an e-commerce environment.

Customer Satisfaction Monitoring

Regular sentiment analysis enables businesses to monitor customer satisfaction and identify changes in customer perception over time.

Product Improvement

Frequently occurring negative words in customer reviews help identify recurring product issues. Manufacturers and sellers can use this feedback to improve product quality and reduce customer complaints.

Customer Experience Enhancement

Customer reviews reveal valuable information about packaging, delivery, product performance, and overall shopping experience. Addressing these concerns can improve customer loyalty and satisfaction.

Marketing Strategy

Products receiving consistently positive reviews can be highlighted in promotional campaigns. Positive customer feedback enhances credibility and influences purchasing decisions.

Decision Support

Automated sentiment analysis enables organizations to process large volumes of customer feedback efficiently, reducing manual effort while providing timely insights for business decision-making.

6.5 Recommendations

Based on the analysis conducted in this project, the following recommendations are proposed:

Improve Product Quality

Products receiving repeated negative feedback should be reviewed to identify manufacturing or quality-related issues.

Monitor Customer Reviews Continuously

Businesses should regularly analyze customer reviews to detect emerging issues before they affect overall customer satisfaction.

Encourage Detailed Customer Feedback

Customers should be encouraged to provide detailed reviews, as they offer richer information than numerical ratings alone.

Enhance Delivery and Packaging

Where reviews indicate concerns regarding shipping or packaging, businesses should collaborate with logistics partners to improve the customer experience.

Implement Automated Feedback Systems

Organizations can integrate sentiment analysis tools into their customer support systems to monitor customer opinions in real time and respond proactively.

6.6 Summary

The results of this project demonstrate that customer review analysis is a valuable technique for understanding customer satisfaction and improving business performance. By combining exploratory data analysis with sentiment analysis, the project successfully transformed unstructured customer reviews into meaningful business insights. The findings emphasize the importance of customer feedback as a strategic resource for

product improvement, customer retention, and data-driven decision-making.

Table 6.1: Summary of Results

Aspect	Observation
Customer Ratings	Predominantly high ratings indicate overall satisfaction.
Customer Sentiment	Positive sentiment was the most frequently observed category.
Rating vs. Sentiment	Numerical ratings and textual sentiment were generally consistent.
Review Characteristics	Most reviews were concise, while negative reviews often contained greater detail.
Business Value	Customer feedback provides actionable insights for product and service improvement.

7: CONCLUSION AND FUTURE SCOPE

7.1 Conclusion

This project successfully explored customer sentiment using the Flipkart Product Reviews Dataset by applying data analytics and Natural Language Processing (NLP) techniques. The dataset was systematically examined, cleaned, and prepared before conducting exploratory data analysis and sentiment classification.

Exploratory Data Analysis provided valuable insights into customer rating patterns, review characteristics, and product popularity. Various visualizations helped identify trends within

the dataset and established a strong foundation for further analysis.

Sentiment analysis was performed using the TextBlob library, which classified customer reviews into Positive, Neutral, and Negative categories based on textual polarity. The comparison between customer ratings and textual sentiment demonstrated that written reviews generally reflected the numerical ratings provided by customers, while also revealing a few interesting inconsistencies.

Overall, the project demonstrates how customer feedback can be transformed into meaningful business intelligence. The insights obtained from this analysis can assist businesses in improving product quality, enhancing customer satisfaction, and making data-driven decisions based on customer opinions.

7.2 Limitations of the Study

Every project has limitations, and acknowledging them is a sign of good analysis rather than a weakness.

Although the project achieved its objectives, certain limitations should be considered:

- The analysis was limited to the provided Flipkart review dataset and did not include real-time customer reviews.
- Sentiment classification was based on TextBlob, which relies on polarity scores and may not accurately interpret sarcasm, idiomatic expressions, or complex contextual meanings.
- The dataset included only textual reviews and numerical ratings. Other factors such as customer demographics,

product categories, pricing, and purchase history were not available for analysis.

- Since the dataset represents a specific collection of reviews, the findings may not be fully representative of all Flipkart customers or products.

7.3 Future Scope

The project can be extended in several ways to improve both analytical depth and practical applicability.

- Apply advanced deep learning models such as BERT, RoBERTa, or LSTM to improve sentiment classification accuracy.
- Analyze customer reviews in multiple languages to support a broader range of users.
- Develop an interactive dashboard using Power BI, Tableau, or Streamlit for real-time visualization of customer sentiment.
- Perform product-category-wise sentiment analysis to identify strengths and weaknesses across different categories.
- Integrate real-time review data from e-commerce platforms to build a live customer sentiment monitoring system.
- Expand the analysis by incorporating additional customer attributes, allowing more detailed segmentation and personalized business insights.

7.4 References

Use a consistent citation style. Since this is an internship report, a simple numbered list is sufficient.

1. Flipkart Product Reviews Dataset (Dataset provided for the Data Analytics Capstone Project).
2. Loria, S. (2024). *TextBlob: Simplified Text Processing*. https://textblob.readthedocs.io/
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